LEF MODEL · CHAPTER 9

Chapter 9: The Clock on the Floor — Melted, Smeared and Sharp Craters, the Feeding Engine and the Three Generations of Exposure

Based on Paper 05 — The Maria (Parts G–I: Studies 8 and 15)

01 · THE MYSTERY

The mystery: why are the large craters in Mare Imbrium melted, the medium ones smeared and the small ones sharp?

Look at the floor of **Mare Imbrium**, the great mare of the near side. On exactly the same uniform basalt surface, three famous craters appear side by side in three completely different states:

- **Sinus Iridum (246 km across):** a giant crater half melted away — only its northern rim survives as the ridge of the Jura Mountains, while its southern rim has been erased completely and its floor has merged with the surface of the mare (a depth-to-diameter ratio of 0.008, only a quarter of an ordinary highland crater of its size!).
- **Plato (101 km across):** a medium crater on the northern shore whose ring of walls is complete, but whose floor is filled, flat and completely smeared (a depth-to-diameter ratio of 0.020, about half that of an ordinary highland crater).
- **Kirch (11 km across):** a small crater in the heart of the mare floor that has remained sharp, crisp and even deeper than a fresh highland crater (0.166 compared with 0.078)!

All three sat on exactly the same mare, at the same latitude and on the same ground. If they had all been created by random asteroid impacts over billions of years, why is there not even a single sharp crater 20 km or more across on the floor of Imbrium, and why does the state of preservation of the craters follow their size so precisely?

02 · AN IMAGE

An image: a spoon pressed into a pot of cooling porridge — and an open tap on a frosty night

Two simple images reveal the physical mechanism that links the size of the craters, the cooling of the maria and the order of their exposure:

- **A pot of scalding porridge that cools slowly (the three crater states):** take a pot of thick porridge taken off the heat, and press a spoon into it at three stages. While the porridge is still scalding and fluid (low viscosity, about 1 Pa·s), the dent closes and vanishes at once — only a faint mark on the rim remains (**a melted crater** like Sinus Iridum). When the porridge cools to the texture of a thick "mud" full of crystals (mush), the dent keeps its ring of rim but its floor slumps and fills (**a smeared crater** like Plato). And when the porridge is frozen and completely solid (below the solidus, ~ 1,050°C), the press of the spoon leaves a sharp, clear dimple forever (**a sharp crater** like Kirch)!
- **A running water tap on a frosty night (the critical flux of the feeding engine):** why did some openings freeze early while others stayed open to the end? Imagine a water pipe outdoors on a frosty night. As long as water flows through it strongly, the heat it carries prevents the pipe from freezing; as soon as the flow weakens below a certain threshold (**the critical flux**), the cold wins and the pipe is sealed with ice!

03 · SCIENCE AND MODEL

Science and model: two hands of one clock — the cooling of the maria and the waning of the firing from

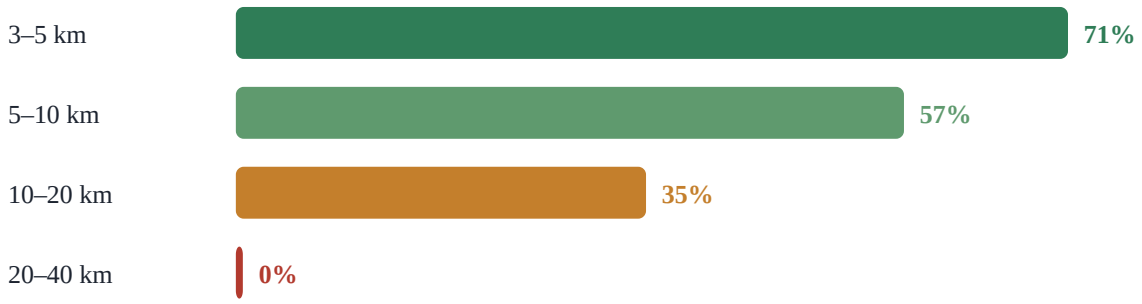
the Earth

Part G of Paper 05 presents a joint study with Paper 06 (**Study 8**), which scanned every crater 3 km or more across on the near-side basalt (3,478 craters in all: 324 on the mare floors above the feeding domes, and 3,154 on the basalt plains). Each crater was compared with highland craters of exactly the same diameter, and classified as **sharp**, **smeared** (shallower than 75% of the highland craters) or **melted** (shallower than 95% of them):

1. **The craters that vanished in the mare floors:** on the floors of the contact maria (Imbrium and Serenitatis) there are only a third of the 2–5-km craters found on the plains beside them (263 against 836 per million km²), while for craters of 20 km and more the number is **absolutely zero (0)**! Every large crater that landed on them while they were fluid was simply swallowed and erased.

2. **The clear link between a crater's size and its state:** on the mare floors, the share of **sharp** craters drops from **71%** at 3–5 km, to 57% at 5–10 km, to 35% at 10–20 km, and reaches **0%** at 20–40 km — while the share of **melted** craters leaps from 17% to **85%** and to **100%** above 80 km (statistical significance better than one in ten billion!).

3. **Why did the large ones arrive first and the small ones last? (§G.4–G.5):** after the detachment, the Earth continued to fire glowing material into space. As the Moon receded from the Earth, the minimum speed needed to reach it rose ($v_{\min} = \sqrt{2GM_{\oplus}(1/R_{\oplus} - 1/r)}$ — from 8.8 km/s at a distance of 17,000 km to 10.6 km/s at 60,000 km). At the same time, the magma storm on the Earth gradually calmed: first large glowing lumps were fired (when the Moon was close and the maria were fluid → melted craters), then medium lumps (when the maria were in the "mush" state → smeared craters), and finally small glowing drops (when the maria had already frozen → sharp craters, for example 82% of the 3–10-km craters are sharp in the contact maria!).



The share of sharp craters on the mare floors by diameter (Study 8): the larger the crater, the earlier it struck a fluid floor.

The Earth's heat from below — the proof from the crater rims on the contact plains (Study 15, §G.6)

The accepted literature holds that the large craters on the plains (such as Oceanus Procellarum) were flooded by lava from the side — like a sandcastle around which water rises and covers its lower slopes, but leaves the crest of the rim intact. In the LEF model, the contact plains were heated **from below** by the radiation of the nearby Earth (T ≈ 2,000 K, view factor $0.205 \rightarrow T_{\text{surface}} \approx 1,345\ \text{K}$, above the melting point of basalt!).
Examining 36 radial profiles around every crater 40 km or more across reveals the difference: on the contact plains, the rim **disappears completely in +24% more directions** facing the plain than toward the highland (compared with **0%** on the plains beyond the contact!), and the surviving crest is much blunter and more worn (curvature lower by −35 m/km², width greater by +0.14 km and angle wider by +5.3°) — thermal wear from below that side flooding can never produce!

04 · THE MECHANISM IN WORDS

The mechanism in words: the feeding engine, the exposure front and the three generations of openings

Parts H and I of Paper 05 show how the physics of the feeding engine (pressure, buoyancy, viscosity and volatiles) connects with the geometry of the shrinking

contact disc (from 6,000 km to 3,000 km) and creates **three generations of openings** on the face of the Moon:

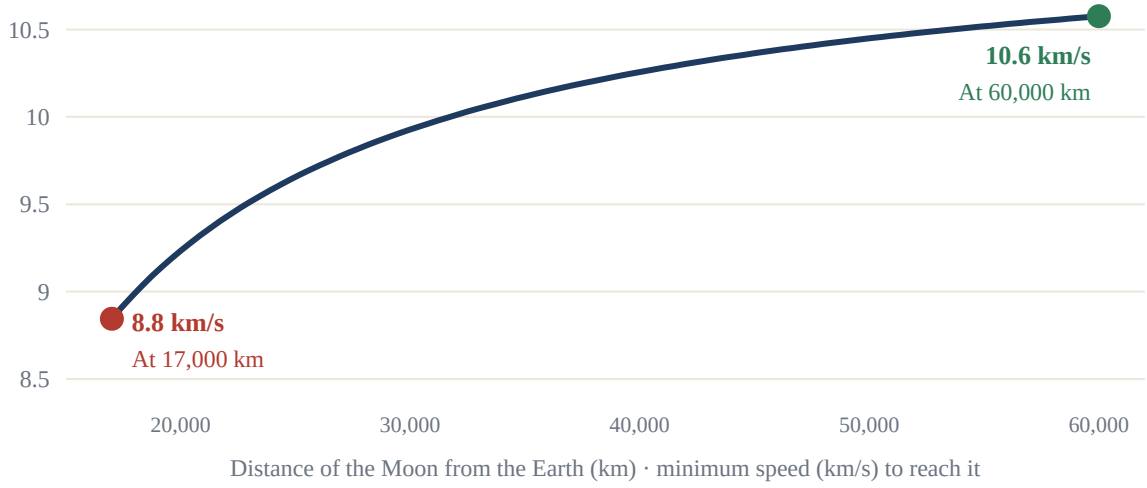
1. **Why only a fluid can transmit the pressure (§H.1–H.2):** a solid in the shell would have had to carry a stress of 13–16 GPa (60–80 times its strength, ~ 0.2 GPa) and would have broken at once. So the opening is a column of fluid magma that transmits hydrostatic pressure and buoyancy, with the content of water and volatiles setting its viscosity: low enough to flow freely beneath the stiff shell, yet high enough to carry heavy iron grains with it without their sinking on the way.

2. **The link between the feeding flux and the closing of the opening (§H.3):** as long as the rising flux ($Q_{\text{feed}} = Q_{\text{retained}} + Q_{\text{erupted}}$) brings heat at a rate exceeding the rate of cooling to space, the opening stays open. When the flux falls below the threshold of the **critical flux**, the opening freezes and seals (t_{shutdown}). On the near side, the radiation of the glowing Earth (~ 2,000 K, view factor 0.205) kept the surface at a temperature of about **1,345 K** — above the melting point of basalt (~ 1,320 K or 1,050°C) — so the openings of the near side **could not freeze as long as the Moon was attached**! On the far side, by contrast, as the pear lengthened and less heat from the Earth reached it, the openings froze one after another (except **Moscoviense**, which was fed from Mare Serenitatis through a straight internal channel).

3. **The moving exposure front and the three generations of openings (§I.1–I.5):** when the contact disc shrank from 6,000 km to 3,000 km during the outward rounding, the edge of the contact retreated inward as an **exposure front**:
    - 4. **Generation I (the upper dome — the far side, beyond 70°):** openings that were exposed to space from the first moment, fired material outward that built their edges and the craters of the far side, froze early when the heat diminished, and were buried beneath the later building layers (36 buried domes).
    - 5. **Generation II (the peeling ring — the 31°–70° belt, the hill):** openings that began their lives as **covered mouths** that passed material inward, and when the exposure front passed over them

became **exit openings** that launched material outward for a long time.

- 6 **Generation III (the contact disc — the near cap up to 31°):** the openings that stayed covered and pressed against the Earth right up to the moment of detachment (Imbrium, Serenitatis, Copernicus H and Sinus Aestuum). They served as feeding mouths for the Earth's pressure into the Moon; at the moment of detachment the pressure stopped all at once, the eruptions halted, and their deep, fluid floors froze slowly from the outside inward over hundreds of millions of years!



As the Moon receded, a higher launch speed from the Earth was needed to reach it — and the firing weakened with time: large at first, small at the end.

05 · THE KEY EQUATIONS

Academic explanation and the key equations

The cooling of the basalt lid freezing from the top down (the square-root-of-time law, §G.3) and the minimum escape speed needed by material fired from the Earth to reach the receding Moon (§G.4) are expressed as follows:

$$\delta(t) \approx 2\lambda\sqrt{\kappa\, t} \quad \Longrightarrow \quad t_{\text{freeze}} \propto \frac{h^2}{\kappa}$$

$$v_{\min}(r) = \sqrt{2GM_{\oplus}\left(\frac{1}{R_{\oplus}} - \frac{1}{r}\right)}$$

$$t_{\text{shutdown}} \geq t_{\text{exp}}$$

Since the freezing time (t_{freeze}) grows with **the square of the depth (h²)** of the magma reservoir, the deep, hot contact maria (Imbrium and Serenitatis) stayed fluid tens to hundreds of times longer than the shallow openings of the margins — and the condition $t_{\text{shutdown}} \geq t_{\text{exp}}$ establishes that only an opening still active when the exposure front passed over it became an active exit opening.

06 · THE SIGNATURE

The signature in reality: the size of the opening as a memory of the length of feeding

One of the direct geometric predictions of the feeding engine (§I.6) is that a prolonged flow of hot magma under pressure gradually widens the neck of the opening — like running water widening a channel in ice ($D = F(\text{feeding history}, P, T, Q, \eta)$):

- **A clear hierarchy in the diameters of the feeding domes:** in the catalogue of 34 maria (Appendix A of Paper 05), **all six of the largest feeding domes on the Moon (450–909 km) lie on the near side**, and the median diameter of a near-side dome is **375 km** compared with only **227 km** on the far side (where the largest dome is Orientale, 435 km)!
- **The scarcity of large craters on the smooth floors:** on the floor of Mare Imbrium there are only 11 craters 20 km or more across per million km² (and on Serenitatis 16), compared with **205** per million km² on the far-side highlands — a gap of 13–18 times, showing that the contact floors froze only after most of the storm of impacts had already passed.

The lab box: the preservation states of the craters, the wear of the rims and the generations of exposure (Paper 05, Parts G–I)

Study / measure	Comparison and sample	Quantitative finding	Significance and physical meaning
Crater state by diameter (Study 8)	**324 craters (≥ 3 km) on the mare floors**	**Sharp: 71% (3–5 km) → 0% (20–40 km)** **Melted: 17% → 85% (100% above 80 km)**	**Cochran–Armitage trend test: $p < 10^{-10}$; the large craters struck early on a fluid floor**
Craters that vanished in the floors (Study 8)	Mare floors against basalt plains (without rays)	In the contact (2–5 km): 263 against 836 per million km² In the contact (20+ km): **0** against 23 per million km²	Early impacts were swallowed and erased in the fluid magma of the contact maria
Wear of rims on the contact plains (Study 15)	**137 craters (≥ 40 km) in 36 radial profiles**	**Rim missing on the plain side: +24% in the contact against 0% beyond it; crest blunter by −35 m/km² and angle +5.3°**	**The contact plains were melted and heated from below by the Earth (1,345 K), not merely flooded from the side**
The three generations of exposure (§I.4)	The geometry of the retreating contact ($6,000 \rightarrow 3,000$ km)	Generation I (>70°): exposed at once and closed early Generation II (31°–70°): a mouth that became an exit opening Generation III (<31°): contact mouths that closed at detachment	The order of cooling: far side → belt → contact zone (Imbrium and Serenitatis)
Feeding-dome diameter as memory (§I.6)	The 34 maria in the Appendix A catalogue	Near-side median: **375 km** (up to 909 km) Far-side median: **227 km** (up to 435 km)	Prolonged feeding in the high heat of the near side widened the feeding openings

07 · NEXT

The door ahead

The craters on the mare floors have revealed the exact timetable of the cooling: when the Moon was close and the maria were fluid, large lumps landed on them and were swallowed or melted into rings; when they turned into crystal "mush",

the medium craters were smeared; and only when they had frozen completely did the last small drops from the Earth leave sharp craters.

But to prove once and for all that the maria are indeed **the source** that fired material into their surroundings, and that the contact zone is where the Moon held on to the Earth until the last moment, we must check the chemical fingerprint of the rocks. In the next chapter — **Chapter 10: The Chemical Fingerprint — the Iron Rings, the Thorium Riddle in the Contact Zone and the Five Clocks** (from Paper 05 — *The Maria*, Parts J–O and Appendix A) — we will close the file on the maria of the Moon!

Chapter 10

The Chemical Fingerprint — the Iron Rings, the Thorium Riddle in the Contact Zone and the Five Clocks →

← Chapter 8 · Prologue

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